AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Mathematics (Science Group) Paper I

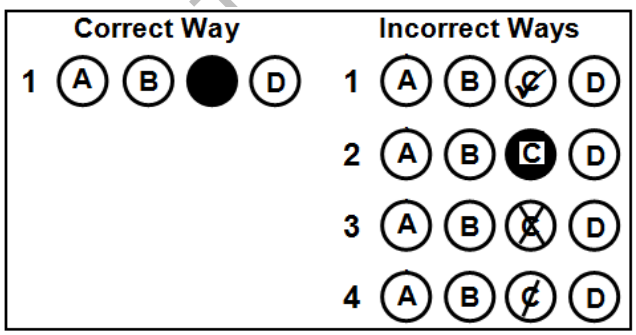
Time: 1 hour and 20 minutes Marks: 45

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

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INSTRUCTIONS:

1. Check your name and school information on the question paper and Optical Mark Recognition (OMR) sheet. Sign on both if the information is correct.
2. Read each question carefully.
3. Answer all questions on the provided OMR sheet within the allotted time. DO NOT mark your answers on the question paper.
4. There are 100 answer numbers on the OMR sheet. Use answer numbers 1 to 45 only.
5. Each question has four choices: A, B, C, and D. Choose ONE correct answer only. On the OMR sheet, completely fill in the circle corresponding to your choice with a pencil as shown in the example below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with an eraser, before filling in the new circle. Ensure your marks are dark and clear. Do not make stray marks or leave incomplete fillings as the OMR machine ONLY records what is in the circles.
7. Do not fold, tear, or damage the OMR sheet. Damaged sheets may be scanned incorrectly.
8. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
9. You may use a simple calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meaning.
- The same formulae will be provided in the annual and september examinations.

Real Numbers

$x^m \times x^n = x^{m+n}$ $\quad (x \times y)^n = x^n \times y^n$ $\quad (x^m)^n = x^{mn}$

$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$, where $y \neq 0$ $\quad \frac{x^m}{x^n} = x^{m-n}$

Logarithms

$\log_a(m \times n) = \log_a m + \log_a n$ $\quad \log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$ $\quad \log_a b = n \Leftrightarrow a^n = b$

$\log_a(m)^n = n \log_a m$ $\quad \log_m n = \frac{\log_a n}{\log_a m}$

Sets Theory

$A \Delta B = (A \cup B) - (A \cap B)$ $\quad (A \cup B)^c = A^c \cap B^c$ $\quad (A \cap B)^c = A^c \cup B^c$

Factorisation

$(a+b)^2 = a^2 + 2ab + b^2$ $\quad (a-b)^2 = a^2 - 2ab + b^2$

$(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$ $\quad (a+b)^2 - (a-b)^2 = 4ab$

$(a+b)(a-b) = a^2 - b^2$ $\quad (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ $\quad (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ $\quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Introduction to Coordinate Geometry

$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $\quad \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$y = mx + c$ $\quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Similar Figures

$\frac{1}{2} h \times b$ $\quad l \times b$ $\quad l^2$ $\quad 6l^2$

l^3 $\quad \pi r^2$ $\quad l \times b \times h$ $\quad 2(l \times b + b \times h + l \times h)$

Introduction to Trigonometry

$1° = \frac{\pi}{180} rad$ $\quad 1rad = \left(\frac{180}{\pi}\right)°$ $\quad H^2 = P^2 + B^2$

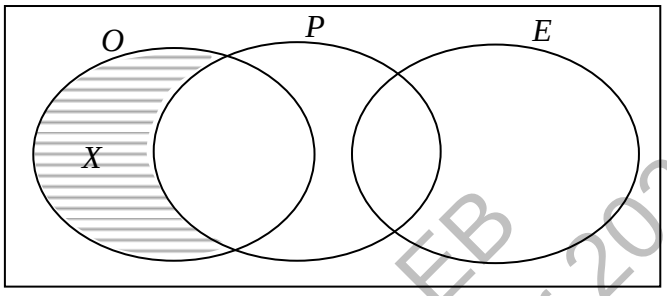
$\sin^2\theta + \cos^2\theta = 1$ $\quad 1 + \tan^2\theta = \sec^2\theta$ $\quad 1 + \cot^2\theta = \text{cosec}^2\theta$

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1. Given that $A = \{1, 2\}$ and $B = \{1, \{2, 3\}\}$, then $A \cap B$ will be

   A. {1}.
   B. {1, 2}.
   C. {1, {2}}.
   D. {{1, 2}}.

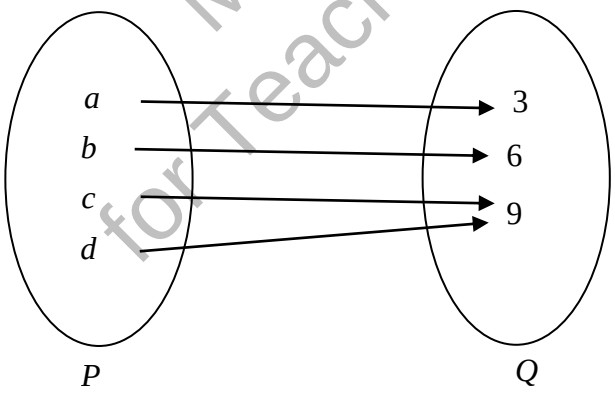
2. There are 10 balls in a bag. The balls are numbered as 1 to 10. Amaan wants to separate the balls in three sets of even (*E*), odd (*O*) and prime (*P*) numbers as shown in the given Venn diagram.



   The shaded region, represented by *X*, will contain the balls numbered as

   A. $\{2, 4, 6, 8, 10\}$.
   B. $\{1, 3, 5, 7, 9\}$.
   C. $\{1, 3, 5, 7\}$.
   D. $\{1, 9\}$.

3. The domain of the binary relation, between set *P* and set *Q*, in the given diagram is



   A. {*a*, *b*, *c*, *d*}.
   B. {*a*, *b*, *d*}.
   C. {*a*, *b*, *c*}.
   D. {*c*, *d*}.

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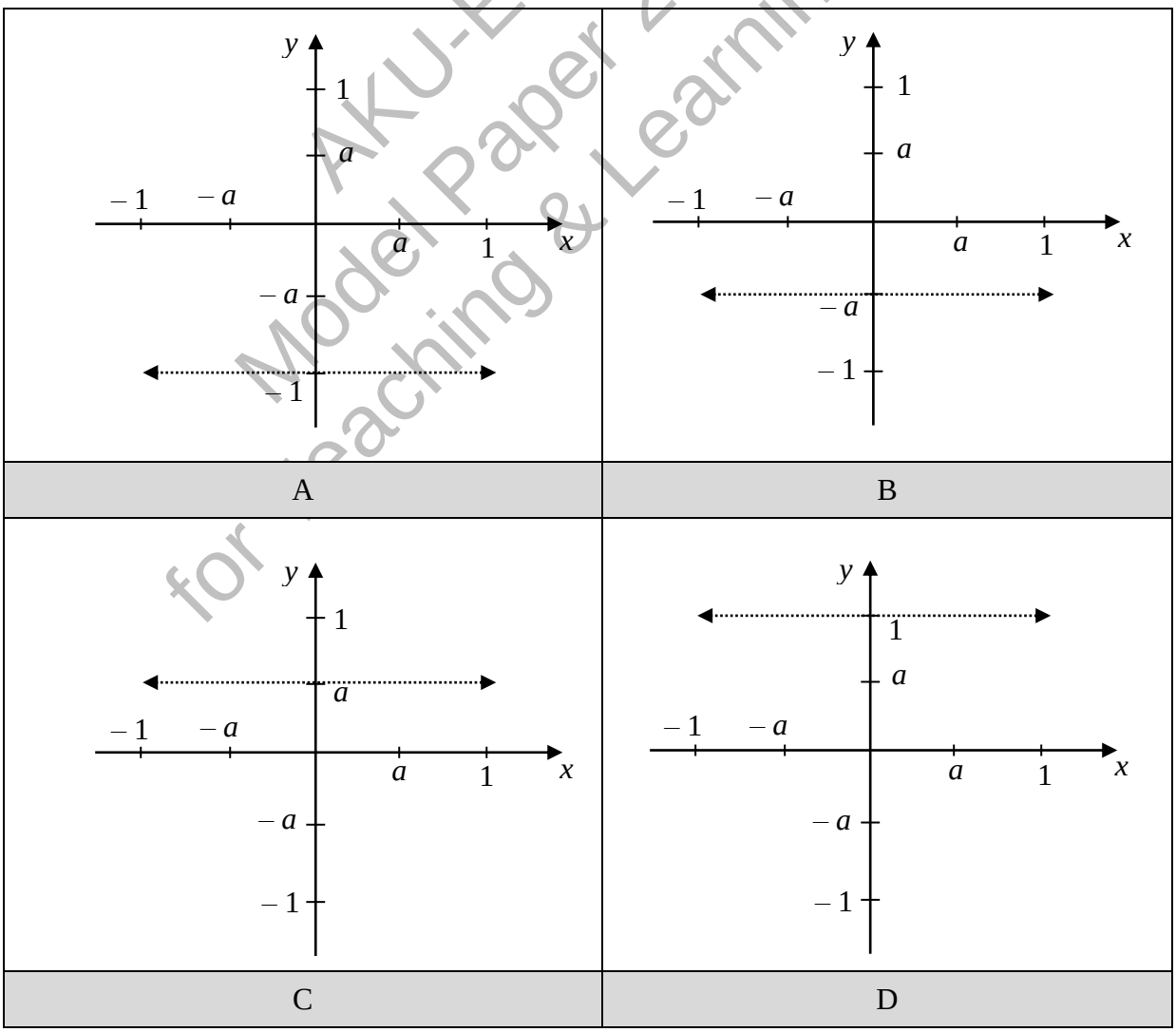
4. If $A = \{1, 2, 3, 4\}$ and $B = \{10, 20\}$, then which of the following is an into function from A to B?

   A. $\{(1, 10), (2, 20)\}$
   B. $\{(1, 10), (2, 10), (3, 10)\}$
   C. $\{(1, 10), (2, 10), (3, 10), (4, 10)\}$
   D. $\{(1, 10), (2, 10), (3, 20), (4, 20)\}$

5. If $A = \{a, e, i\}$ and $B = \{1, 2, 3, 4\}$, then one of the functions from A to B is

   A. $\{(a,1),(e,2)\}$.
   B. $\{(a,1),(e,1),(i,1)\}$.
   C. $\{(a,1),(e,2),(i,3),(a,4)\}$.
   D. $\{(a,1),(a,2),(a,3),(a,4)\}$.

6. The correct graphical representation of $ay + a = 0$ is



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7. $\frac{a^4 - 4a^2}{a^2 - 2a}$ is equal to

A. $a^2 + 2a.$
B. $a^2 - 2a.$
C. $a^2 + 4a.$
D. $a^2 - 4a.$

8. $(a + 2b + c)^2$ is equal to

A. $a^2 + 2b^2 + c^2 + 2ab + 2bc + 2ca.$
B. $a^2 + 4b^2 + c^2 + 4ab + 2bc + 2ca.$
C. $a^2 + 4b^2 + c^2 + 4ab + 4bc + 2ca.$
D. $a^2 + 2b^2 + c^2 + 2ab + 4bc + 2ca.$

9. $u^3 - 27v^3$ can also be expressed as

A. $(u + 3v)(u^2 - uv + v^2).$
B. $(u - 3v)(u^2 + uv + v^2).$
C. $(u + 3v)(u^2 - 3uv + 9v^2).$
D. $(u - 3v)(u^2 + 3uv + 9v^2).$

10. On factorisation of $25x^2 + 5ax + 10x + 2a$, we get

A. $(5x + a)(5x + 2).$
B. $(5a + x)(5a + 2).$
C. $(5x^2 + a)(5x + 2).$
D. $(5ax + ax)(5ax + 2).$

11. $8p^3 + 12p^2q + 6pq^2 + q^3$ can also be expressed as

A. $(2p)^3 + q^3$
B. $(2p + q)^3$
C. $(p + 2q)^3$
D. $p^3 + 2q^3$

12. When $x^2 + kx - 1$ is divided by $x - 1$, the remainder is 2. The value of k is

A. 2
B. 1
C. 0
D. -2

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13. $(t+2)^3$ can also be expressed as

A. $t^3+6t^2+12t+8$
B. $t^3-6t^2+12t-8$
C. t^3+3t^2+3t+8
D. t^3-3t^2+3t-8

14. On simplification of $\left(1-\frac{1}{x}\right)\div\frac{1}{x}$, we get

A. 1
B. $x-1$
C. $\frac{x-1}{x^2}$
D. $\frac{1-x}{x^2}$

15. $45x^2$ is the least common multiple (LCM) of

A. $3x^2$ and $15x$.
B. $3x$ and $15x$.
C. $5x^2$ and $9x$.
D. $5x$ and $9x$.

16. $\frac{b+c}{2}-\frac{b+c}{3}$ is equal to

A. $-\frac{b+c}{6}$
B. $-\frac{b+5c}{6}$
C. $\frac{b+c}{6}$
D. $\frac{b+5c}{6}$

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17. $\sqrt{a^2+\frac{1}{a^2}-2}$ is equal to

A. $\left(a+\frac{1}{a}\right)$.

B. $\left(a-\frac{1}{a}\right)$.

C. $\left(a^2+\frac{1}{a^2}\right)$.

D. $\left(a^2-\frac{1}{a^2}\right)$.

18. The solution set of $\frac{x-1}{2}=0$ is

A. $\{-1\}$.

B. $\{-3\}$.

C. $\{3\}$.

D. $\{1\}$.

19. Given that $|-x|=-1$, the solution set which must satisfy the equation is

A. $\{-1\}$.

B. $\{\ \}$.

C. $\{1\}$.

D. $\{\phi\}$.

20. The inequality which describes $m(x-1)>m$, where m is a natural number, will be

A. $x<1$

B. $x<2$

C. $x>1$

D. $x>2$

21. If x is a negative number, then the mathematical statement which must be TRUE is

A. $-x\leq 0$

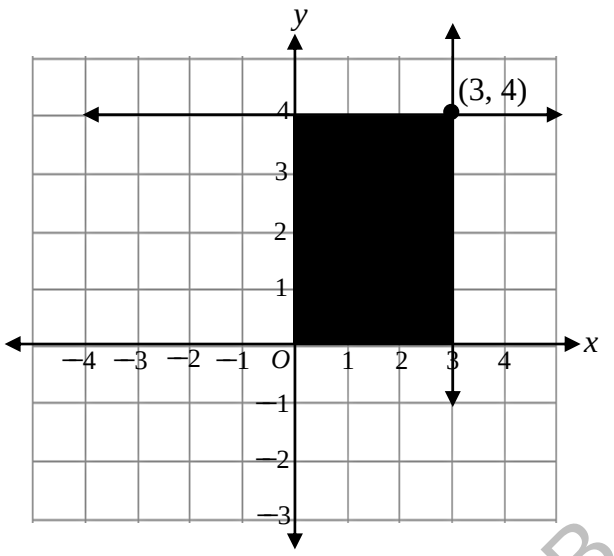
B. $-x\geq 0$

C. $x<0$

D. $x>0$

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22. The given figure shows the solution region of inequalities.



Two of the inequalities satisfying the solution region are

A. $x \leq 4$ and $y \leq 3$
B. $x \leq 3$ and $y \leq 4$
C. $x \geq 4$ and $y \geq 3$
D. $x \geq 3$ and $y \geq 4$

23. The solution set of $\sqrt{x-\frac{1}{4}} = \frac{1}{2}$ will be

A. $\left\{\frac{1}{4}\right\}$.
B. $\left\{\frac{1}{2}\right\}$.
C. $\left\{0, \frac{1}{4}\right\}$.
D. $\left\{0, \frac{1}{2}\right\}$.

24. The solution set of $5 - x < 0$, where $x \in R$, will be

A. $\{x | x \in R,\ x < 5\}$.
B. $\{x | x \in R, x > 5\}$.
C. $\{x | x \in R, x > -5\}$.
D. $\{x | x \in R, x < -5\}$.

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25. The FALSE statement about the rectangular coordinate system in the plane is that the

A. *x*-axis and *y*-axis are parallel to each other.
B. *x* and y components can be positive and equal.
C. ordered pair of real numbers can represent every point in the plane.
D. coordinate system is also known as the Cartesian Coordinate System.

26. If $\vec{MP} = -\left(-3\hat{i}\right)$ and $\vec{MO} = 3\hat{j} + 3\hat{i}$, then the position vector of *P* will be

A. $3\hat{j}$.
B. $-3\hat{j}$.
C. $6\hat{i} + 3\hat{j}$.
D. $6\hat{i} - 3\hat{j}$.

27. The magnitude of the position vector of the point *A* (2, 4, 3) is

A. 9 units.
B. 24 units.
C. $\sqrt{17}$ units.
D. $\sqrt{29}$ units.

28. The valid condition for the distance between two points (a,b) and (c,d) will be

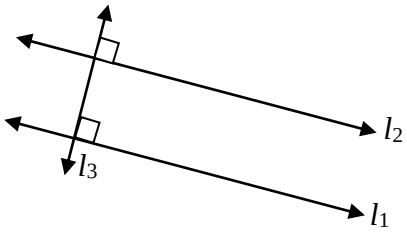
A. $(a-c)^2 + (b-d)^2 > 0$
B. $(b-a)^2 + (d-c)^2 > 0$
C. $(a-c)^2 + (b-d)^2 < 0$
D. $(b-a)^2 + (d-c)^2 < 0$

29. The midpoint of the line segment joining the two points $(2,-2)$ and $(-2,-2)$ is (m,n). The value of n^2 is equal to

A. -2
B. -4
C. 4
D. 0

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30. Consider the given figure.



NOT TO SCALE

If the gradient of the line l_2 is $-\frac{1}{\sqrt{5}}$, then the gradient of the lines l_1 and l_3 will be

A. $-\frac{1}{\sqrt{5}}$ and $-\sqrt{5}$ respectively.

B. $-\frac{1}{\sqrt{5}}$ and $\sqrt{5}$ respectively.

C. $\frac{1}{\sqrt{5}}$ and $-\sqrt{5}$ respectively.

D. $\frac{1}{\sqrt{5}}$ and $\sqrt{5}$ respectively.

31. The straight line that has a slope 2 and y-intercept 8 is

A. $2x - y - 8 = 0$
B. $2x + y + 8 = 0$
C. $-2x - y + 8 = 0$
D. $-2x + y - 8 = 0$

32. The slope-intercept form of the line $3x - 3y + 6 = 0$ will be

A. $y = x + 2$
B. $y = x - 6$
C. $y = 3x + 6$
D. $y = -3x - 2$

33. Consider the two intersecting lines, $y = 2x - 5$ and $y = -2x + 11$.

The locus of points that are equidistant from the given intersected lines are the straight lines

A. $x = 0$ and $y = 0$
B. $x = 4$ and $y = 3$
C. $x = 2$ and $y = -2$
D. $x = 3$ and $y = -4$

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34. Two points, *S* and *T*, are 10 cm apart on a straight line.

The locus of a point *Q* that is equidistant from *S* to *T* is a

A. parallel line to *ST*.
B. point that lies on *ST*.
C. circle with a dimeter *ST*.
D. perpendicular bisector of *ST*.

35. An angle bisector is a locus of a point that is equidistant from two lines.

Which of the following options is/ are correct for the given statement?

I. Parallel lines
II. Perpendicular lines
III. Intersecting lines

A. I only
B. II only
C. I and III
D. II and III

36. Two similar rectangles have sides in the ratio 2:3.

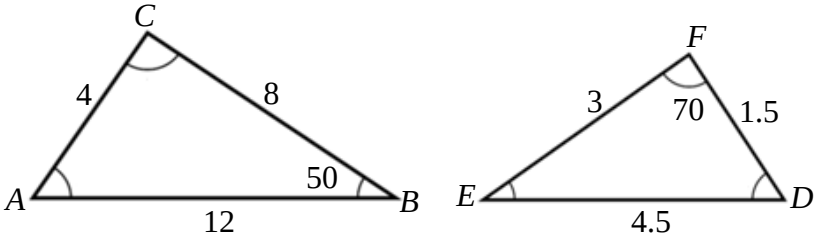
If the area of the larger rectangle is 27 cm², then the area of the smaller rectangle will be

A. 9 cm²
B. 12 cm²
C. 13.5 cm²
D. 18 cm²

37. If two triangles are similar, then their

A. corresponding angles are equal.
B. corresponding sides are equal.
C. perimeters are equal.
D. areas are equal.

38. Sameer has given two triangular buildings to analyse their sizes and angles.



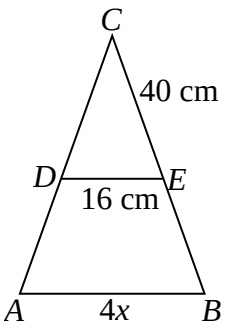
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Based on his analysis the triangles

A. are equal in sides and equal in angles.
B. have equal ratios of sides and are equal in angles.
C. are equal in sides and have equal ratios of angles.
D. have equal ratios of sides and equal ratios of angles.

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39. In the given diagram, *D* and *E* are the midpoints of the sides *AC* and *BC* respectively.



NOT TO SCALE

If $AB = 4x$, $DE = 16$ cm and $EC = 40$ cm, then the value of x is equal to

A. 8 cm.
B. 12 cm.
C. 16 cm.
D. 32 cm.

40. Aliya is designing a square-shaped garden with diagonal pathways measuring 8 m in length. She wants to install a fence around the garden's perimeter to protect the plants.

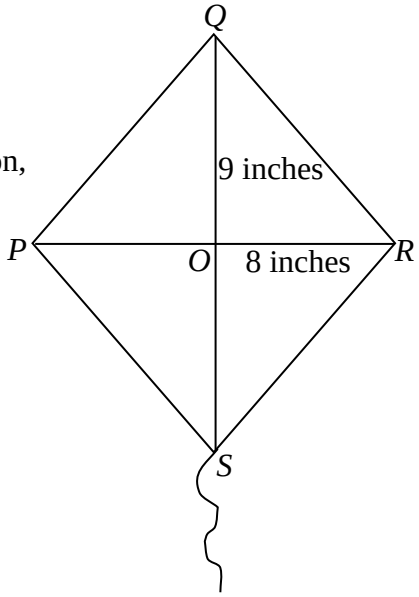
How much fencing material will she need to go all the way around the garden?

A. $4\sqrt{2}$ m
B. $16\sqrt{2}$ m
C. 32 m
D. 48 m

41. Sana is designing a kite for an art competition. The frame of the kite must be a parallelogram, with two bamboo sticks *PR* and *QS* acting as its diagonals as mentioned in a given diagram.

NOT TO SCALE

The sticks are bisecting at point *O*, and the kite's paper is stretched tightly over the frame.



The length of $OR = 8$ inches, $QO = 9$ inches, $PS = (PR - 3)$ inches and $RS = (QS - 4)$ inches.

If she wants to decorate the edges *PQRS* of the kite with a ribbon, then the total length of the ribbon will be

A. 20
B. 27
C. 34
D. 54

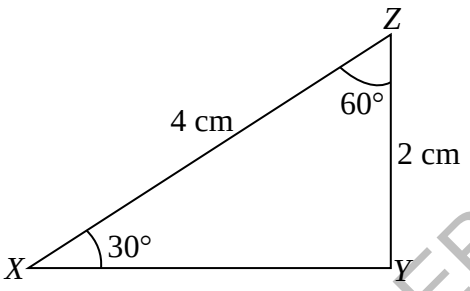
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42. $1\frac{1}{4}$ rotation in anticlockwise direction is equal to

A. 90°
B. 270°
C. 288°
D. 450°

43. In the given triangle XYZ, side XY is equal to

NOT TO SCALE



A. 2 cm.
B. $2\sqrt{3}$ cm.
C. 3 cm.
D. 4 cm.

44. If the length of the sides of a right angled triangle are k, l and m such that $l < k < m$, then according to Pythagoras' theorem,

A. $m^2 = k^2 - l^2$
B. $l^2 = (k+m)^2$
C. $m^2 = (k+l)^2$
D. $l^2 = m^2 - k^2$

45. $\frac{\sin^2\theta}{1-\sin^2\theta}$ is equal to

A. $-tan^2\theta$.
B. $cot^2\theta$.
C. $-cot^2\theta$.
D. $tan^2\theta$.

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